VICTORY 2027

FORCE, WORK, POWER & ENERGY

LECTURE-2 PHYSICS

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Topics To Be Covered

1. COUPLE
2. PRINCIPLE OF MOMENTS
3. CENTER OF GRAVITY
4. QUESTIONS


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SSS ARMY
Where Average
Turns Into Topper
Physics Wallah
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Revision

Motion

- Translational Motion → Force
- Rotational Motion → Torque or Moment of force

$$F = \frac{\Delta p}{\Delta t}$$

$$F = \frac{\Delta(mv)}{\Delta t}$$

$$F = m\frac{\Delta V}{\Delta t}$$

$F = ma$ → Constant

$a > 0$ $v \uparrow$

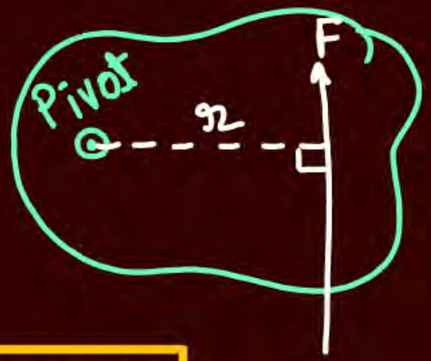
$a < 0$ $v \downarrow$



Torque or Moment of Force

↓

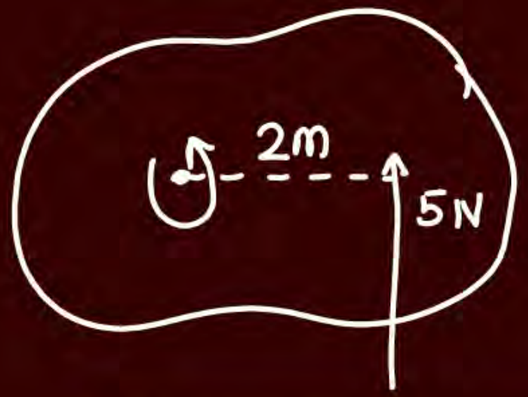
Turning effect of force



$\tau = r \times F$

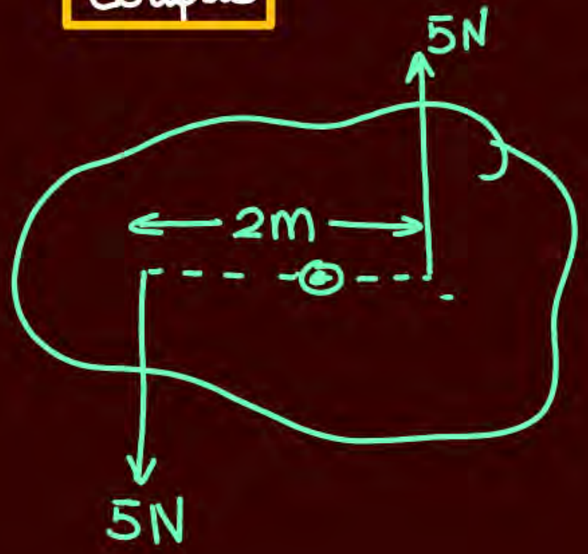
Vector Quantity





$\tau = +2 \times 5 = 10\ Nm$

Couple

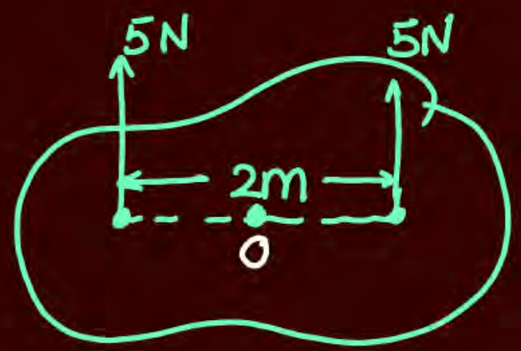


$F_{net} = 0$

$\tau = r \times F$

$= 2 \times 5$

$\tau = 10\ Nm$

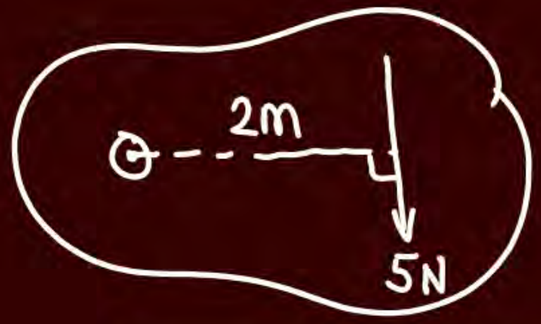


$\tau_1 = +1 \times 5 = 5\ Nm$

$\tau_2 = -1 \times 5 = -5\ Nm$

$\tau_{net} = 0$

$F_{net} = 10N$ (upwards)



$\tau = -2 \times 5 = -10\ Nm$

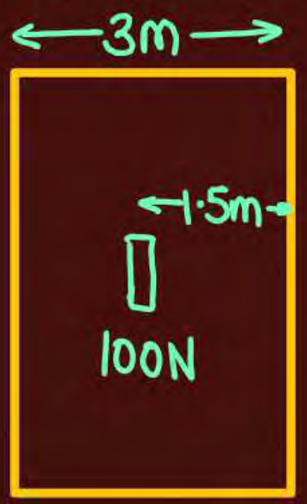
QUESTIONS

The iron door of a building is 3 m broad. It can be opened by applying a force of 100 N normally at the middle of the door. Calculate:

(a) the torque needed to open the door,

(b) the least force and its point of application to open the door.

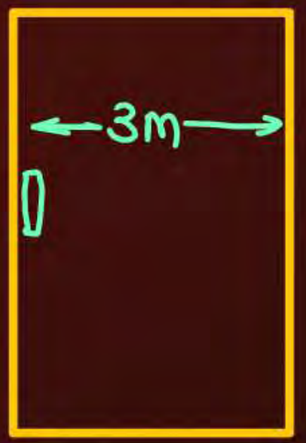
1. 100 Nm & 50 N
2. 150 Nm & 50 N
3. 150 Nm & 25 N
4. 100 Nm & 25 N



Required Torque = $r \times F$

$= 1.5 \times 100$

$= 150$ Nm



$\tau = r \times F$

$150 = 3 \times F$

$F = \frac{150}{3} = 50$ N.

QUESTIONS

The iron door of a building is 3 m broad. It can be opened by applying a force of 100 N normally at the middle of the door. Calculate:

(a) the torque needed to open the door,

(b) the least force and its point of application to open the door.

Solution:

(a) $\mathrm{F} = 100\mathrm{N}$, distance of point of application of force, $r = \frac{1}{2} \times$ breadth of door $= \frac{1}{2} \times 3\mathrm{m} = 1.5\mathrm{m}$

Moment of the force needed to open the door

$$= \mathrm{F} \times r = 100\mathrm{N} \times 1.5\mathrm{m} = 150\mathrm{Nm}$$

(b) The force required will be least if it is applied at the farthest point from the hinges. Therefore the force should be applied at the free end of the door. i.e., at distance of $3\mathrm{m}$ from the hinges. Let the required force be F' newton, then

$$\text{Moment of force } = \mathrm{F}' \times 3\mathrm{Nm}$$

$$\mathrm{F}' \times 3 = 150$$

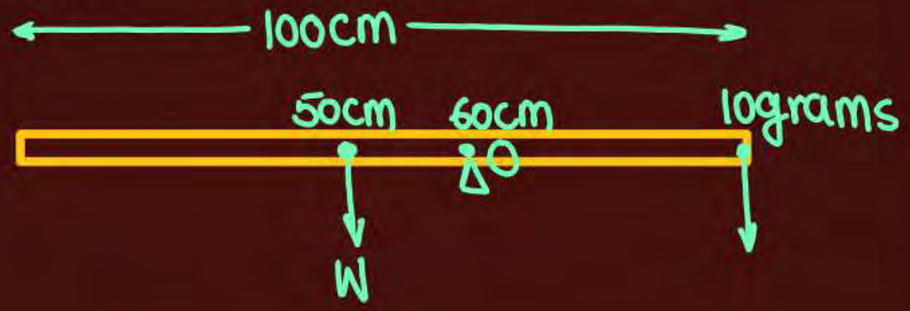
$$\mathrm{F}' = \frac{150}{3} = 50\mathrm{N}$$

QUESTIONS

A uniform metre rule rests horizontally on a knife edge at the 60 cm mark when a mass of 10 g is suspended from one end. Draw diagram of the arrangement.

(a) At which end must this mass be suspended? (opposite to weight end)

(b) What is the mass of the rule?



$\tau_0 = 0$

$-10 \times 40 + m \times 10 = 0$

$-40\phi = -10\,m$

$m = 40g.$

1. 10 g
2. 20 g
3. 30 g
4. 40 g



$W = mg$

Force applied by Earth on object

$g = 9.8\ m/s^2$

$= 980\ cm/s^2$

① $m = 10\ Kg$ $\quad W = mg$

$= 10 \times 9.8$

$= 98\ N$

② $m = 1\ Kg$ $\quad W = mg$

$= 1 \times 9.8$

$1\ Kgf = 9.8\ N$

③ $m = 10g$ $\quad W = mg$

$= 10 \times 980$

$= 9800\ dyne$

④ $m = 1g$ $\quad W = mg$

$= 1 \times 980$

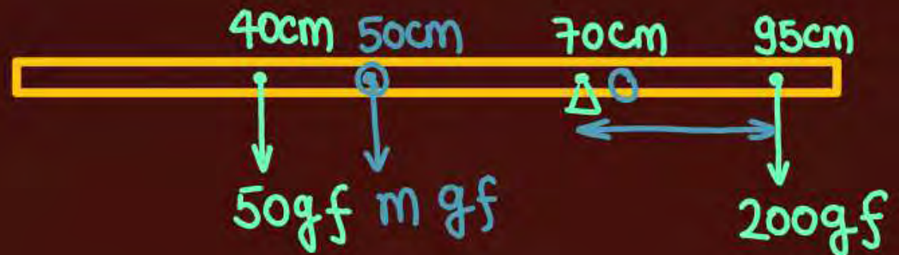
$1gf = 980\ dynes$

QUESTIONS

$\tau_{net} = 0$

A uniform metre rod is balanced at the 70 cm mark by suspending a weight of 50 gf at the 40 cm mark and 200 gf at the 95 cm mark. Draw a diagram of the arrangement and calculate the weight of the metre rod.

1. 150 g
2. 175 g
3. 200 g
4. 225 g



Torque at O = 0

$-25 \times 200 + 20 \times m + 30 \times 50 = 0$

$-5000 + 20m + 1500 = 0$

$-3500 + 20m = 0$

$20m = 3500$

$m = 175\ g$

QUESTIONS

A uniform metre rod is balanced at the 70 cm mark by suspending a weight of 50 gf at the 40 cm mark and 200 gf at the 95 cm mark. Draw a diagram of the arrangement and calculate the weight of the metre rod.



Solution:

Total clockwise moment

$= 200(95 - 70)$

$= 5000\text{gfcm}$

Total anticlockwise moment

$= [50(70 - 40) + \text{m}(70 - 50]\ \text{gf cm} = (1500 + 20\text{m})\ \text{gf cm}$

$\Rightarrow\ 1500 + 20\text{m} = 5000$

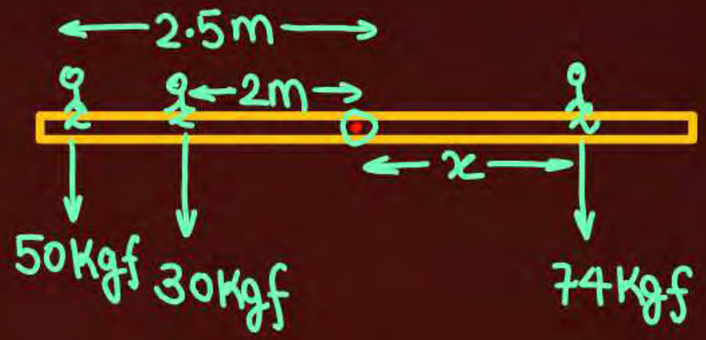
$20\text{m} = 3500$

$\text{m} = 175\text{gm}$

QUESTIONS

On a see-saw, two children of masses 30 kg and 50 kg are sitting on one side of it at distances 2 m and 2.5 m respectively from its middle. Where should a man of mass 74 kg sit to balance it?

1. 2.5 m (on the other side)
2. 2.5 m (on the same side)
3. 3.5 m (on the other side)
4. 3.5 m (on the same side)



$$\tau = 0$$

$$-74x + 30 \times 2 + 50 \times 2.5 = 0$$

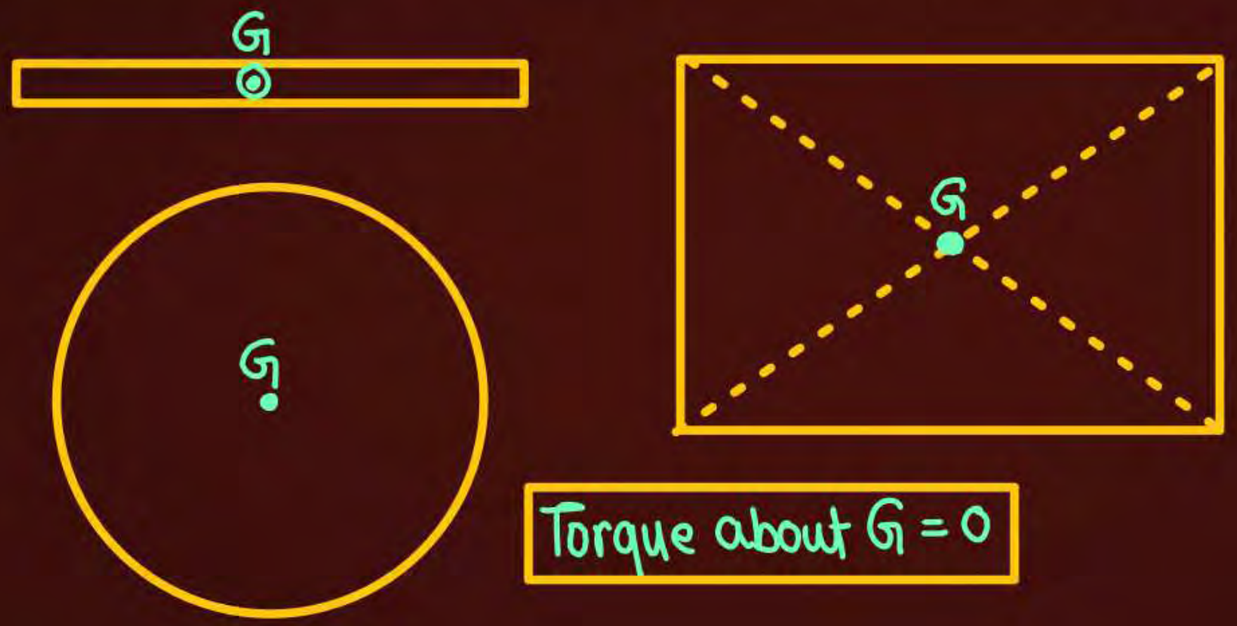
$$-74x + 60 + 125 = 0$$

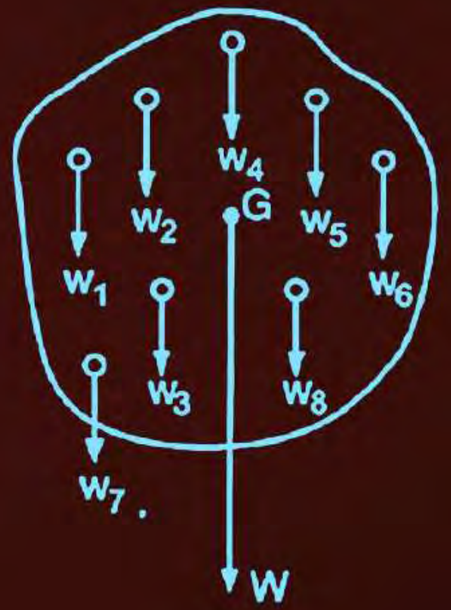
$$-74x + 185 = 0$$

$$x = \frac{185}{74} = 2.5 \text{ m}$$

CENTRE OF GRAVITY

The centre of gravity (C.G.) of a body is the point about which the algebraic sum of moments of weights of all the particles constituting the body is zero. The entire weight of the body can be considered to act at this point, howsoever the body is placed.





CENTRE OF GRAVITY

Note:

(1) The position of the centre of gravity of a body of given mass depends on its shape i.e., on the distribution of mass of particles in it. It changes if the body is deformed.

Example: The centre of gravity of a uniform wire is at the middle of its length. But if the same wire is bent into the form of a circle, its centre of gravity will then be at the centre of the circle.

(2) It is not necessary that the centre of gravity always be within the material of the body.

Example: The centre of gravity of a ring or a hollow sphere lies at its centre where there is no material.

(3) By the concept of centre of gravity, a body of weight W can be considered as a point particle of weight W at its centre of gravity.

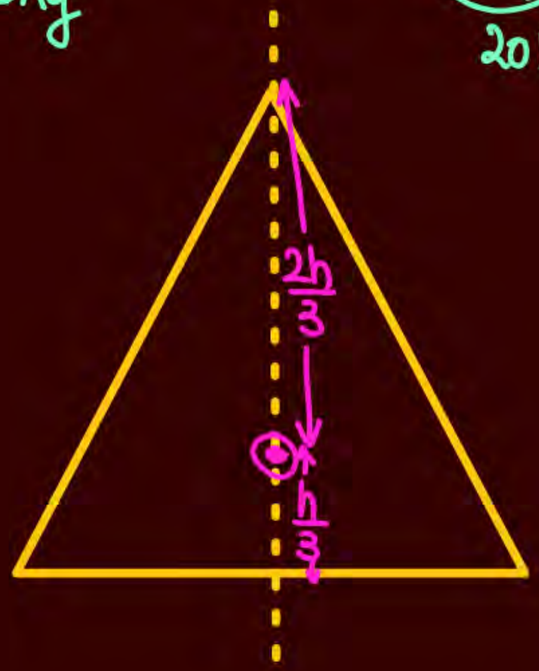


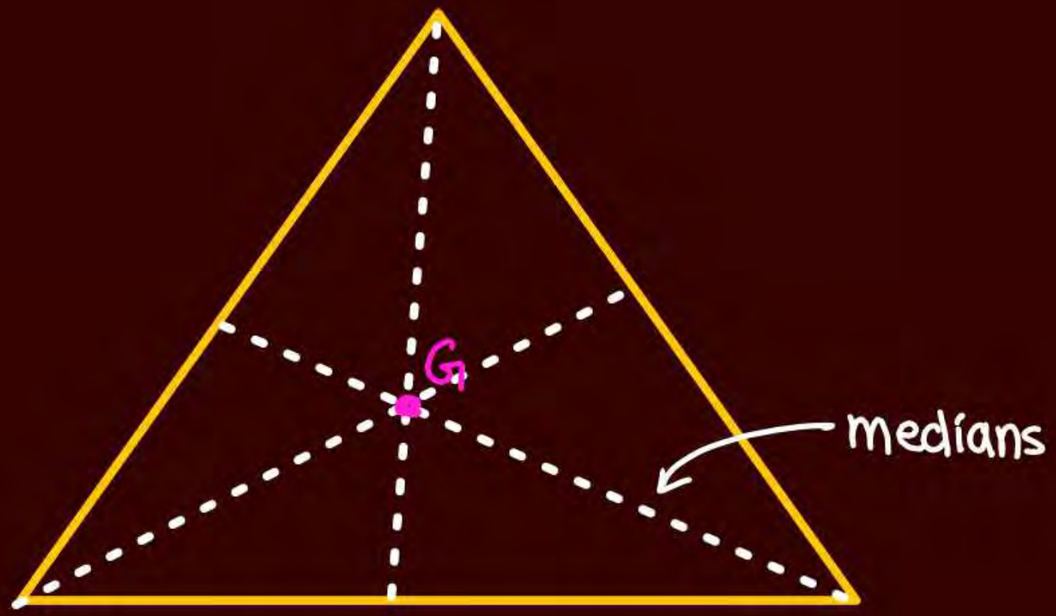




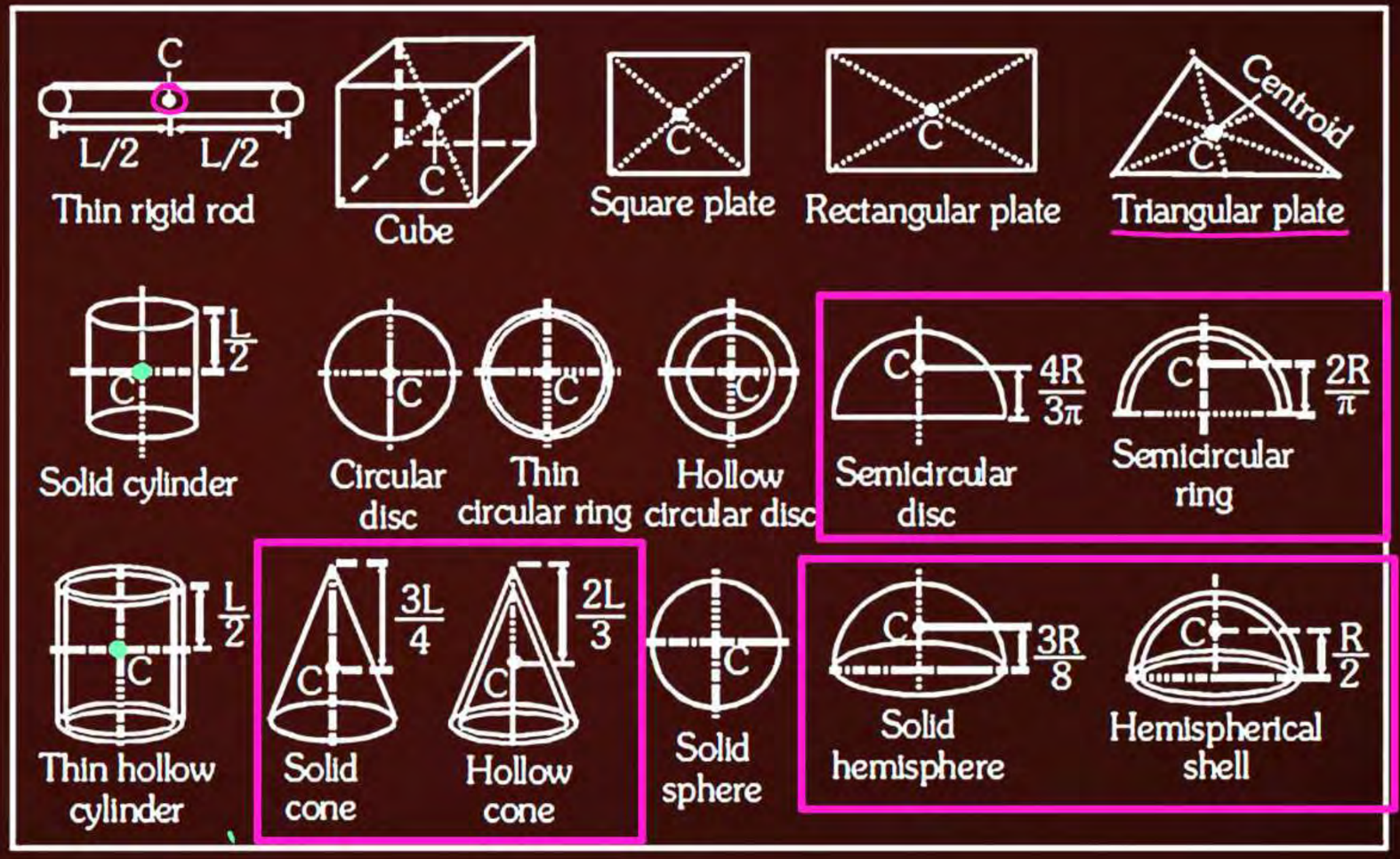
Center of Gravity lies near heavy body






G
medians

CENTRE OF GRAVITY



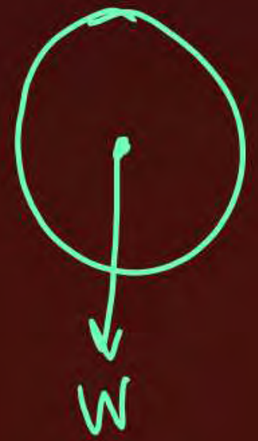
CENTRE OF GRAVITY AND THE BALANCE POINT

A solid body can be balanced by supporting it at its centre of gravity.

QUESTIONS

The centre of gravity of a uniform ball is:

1. at its geometrical centre
2. at its bottom
3. at its topmost point
4. at any point on its surface.

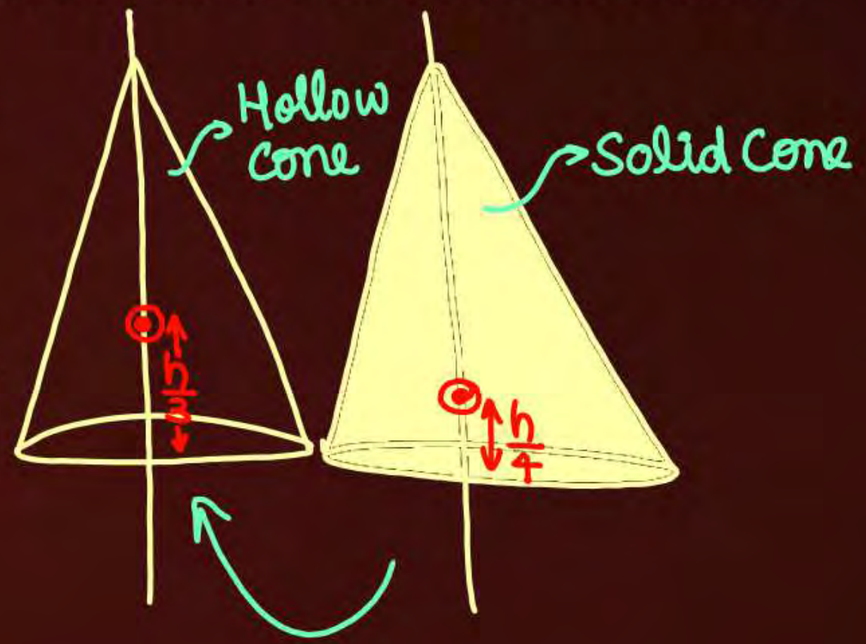


QUESTIONS

The centre of gravity of a hollow cone of height h is at distance x from its vertex where the value of x is :

1. $h/3$
2. $h/4$
3. $2h/3$
4. $3h/4$





QUESTIONS

Centre of gravity is _____ when it is a hollow cone.

1. raised from the base
2. lower towards the base
3. centre of vertical axis
4. reduce from the base

QUESTIONS

The resultant moment of couple, if two equal forces each of 6 N are acting at a distance of 3 cm each from the centre is:

1. 36 Nm
2. 3.6 Nm
3. 360 Nm
4. 0.36 Nm

QUESTIONS

State two methods of increasing the stability of the body.

QUESTIONS

State two methods of increasing the stability of the body.

Solution:

The stability of a body is increased if:

(i) The centre of gravity is as low as possible. It should be above the base and near the geometric centre of the body. This is the case when mass of the body is large at the bottom.

(ii) The area of base of the body is large so that the vertical line through the C.G. of the body passes through its base. If the body is slightly tilted, its C.G. rises higher and it has a tendency to regain its original position to lower the C.G.

HOMEWORK



REVISE THIS LECTURE AGAIN.

PRACTISE 20 QUESTIONS DAILY.

COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

A uniform metre rule rests horizontally on a knife edge at the 60 cm mark when a mass of 10 g is suspended from one end. Draw diagram of the arrangement.

(a) At which end must this mass be suspended?

(b) What is the mass of the rule?

1. 10 g
2. 20 g
3. 30 g
4. 40 g

QUESTIONS

A uniform metre rule rests horizontally on a knife edge at the 60 cm mark when a mass of 10 g is suspended from one end. Draw diagram of the arrangement.

(a) At which end must this mass be suspended?

(b) What is the mass of the rule?

Solution:

From the principle of moments,

Anticlockwise moment = Clockwise moment

$$\mathrm{Mg} \times (60 - 50) = 10\mathrm{g} \times (100 - 60)$$

or $\mathrm{Mg} \times 10 = 10\mathrm{g} \times 40$

$\therefore$ Mass of rule $\mathrm{M} = 40\mathrm{g}$

Thank You